

- 8 The photograph shows a species of fungus growing on a dead tree in a woodland habitat.



fungus

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Wood contains a high proportion of lignin and cellulose.

Fungi can break down molecules in wood by releasing extracellular enzymes.

- (a) Describe the role of the rough endoplasmic reticulum in the formation of extracellular enzymes.

(3)

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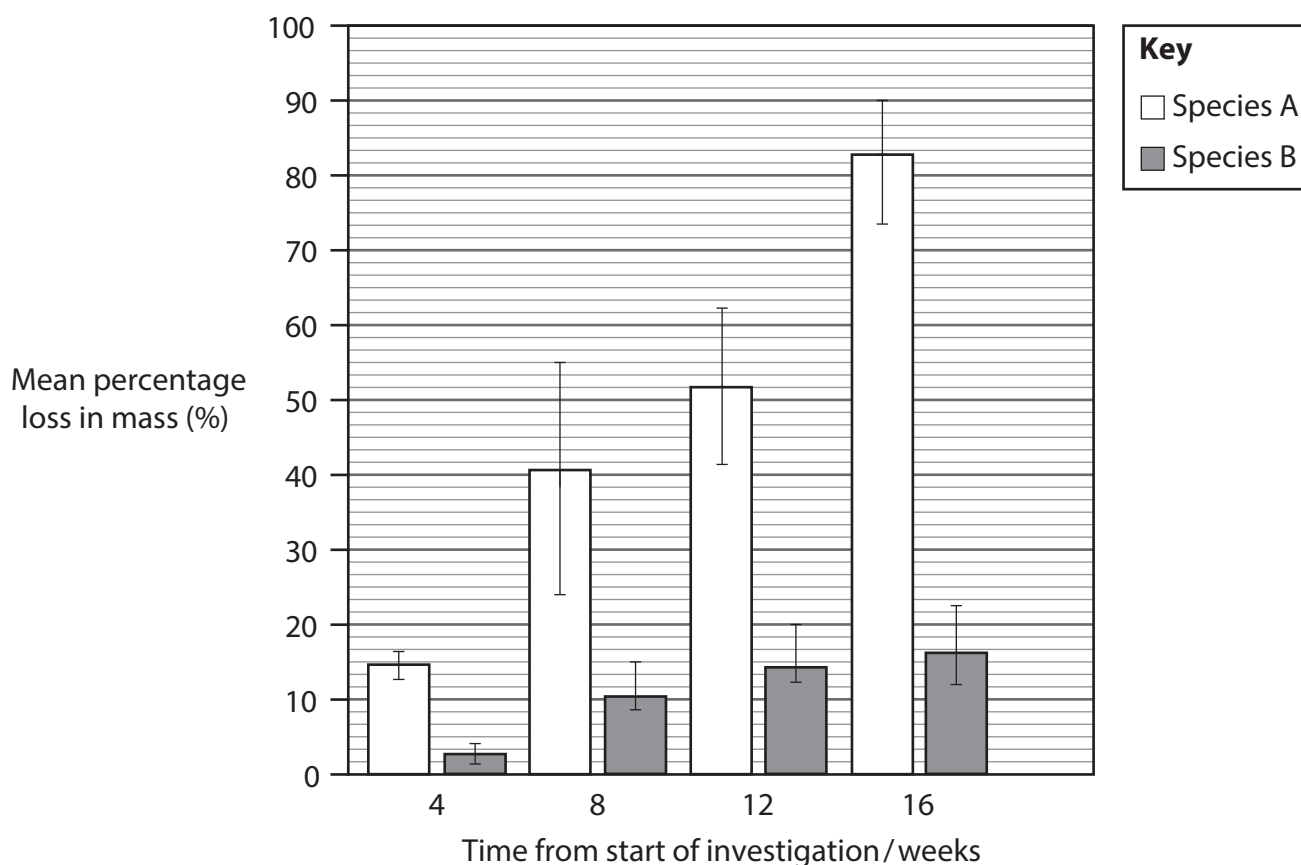
(b) Different species of fungus can break down different molecules in wood.

The effect of two species of fungus on the percentage loss in mass of wooden blocks was investigated.

Species A was grown on one set of wood blocks and species B was grown on another set of wood blocks.

The percentage loss in mass of the blocks was recorded every four weeks.

The graph shows the results of the investigation.



(i) Comment on the results of this investigation.

(3)

(ii) Suggest **one** reason for the difference in breakdown of the wood by species A and species B.

(1)

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(c) A study compared the biodiversity of two woodland habitats.

The index of diversity for habitat one was 4.2.

The table shows the data obtained from habitat two.

Species	Number of individuals (n)	n(n-1)
A	34	1122
B	6	30
C	27	
D	4	12
E	9	72
F	31	930
G	120	14280

(i) An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N-1)}{\sum n(n-1)}$$

Calculate the index of diversity for habitat two.

Use the table and the formula to help you.

(3)

Answer

(ii) State which habitat has the higher biodiversity.

Give a reason for your answer.

(1)

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(iii) Species D is an endangered plant species.

Explain the processes that could be used by a seed bank to conserve this plant species.

(3)

(Total for Question 8 = 14 marks)

TOTAL FOR PAPER = 80 MARKS

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